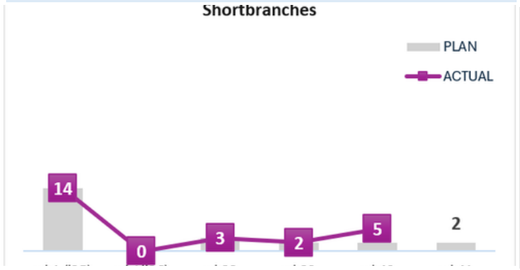
FMDS UPDATE WEEK 40

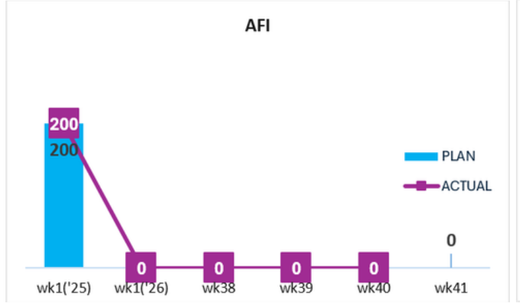
Automation

PPA JUMPER AUDIT WEEK-40
Process Plant Area

Dryer – CM : 6 Jumpers



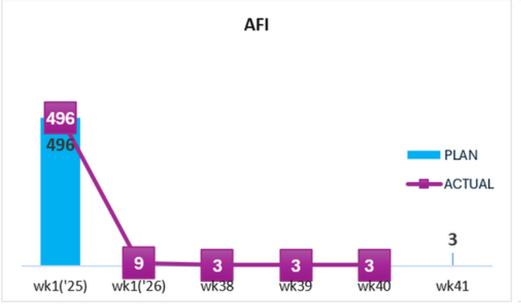




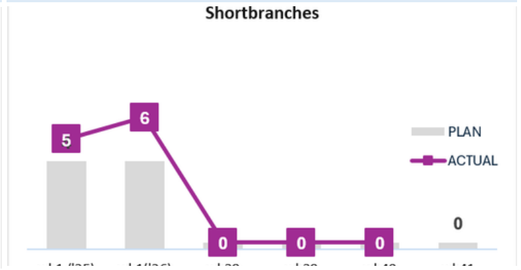
Kiln – CTS : 3 Jumpers

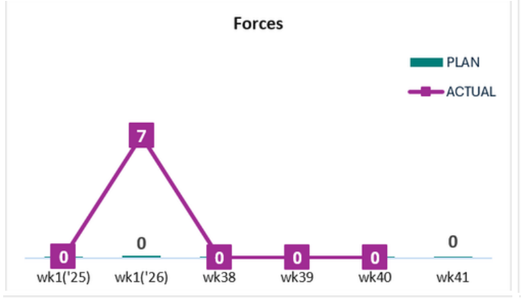


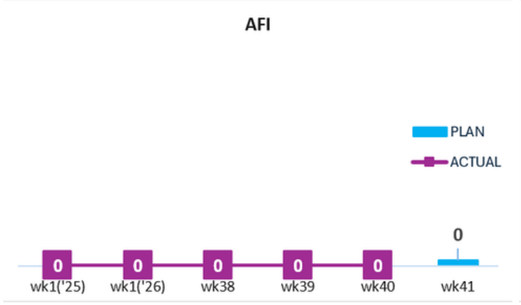




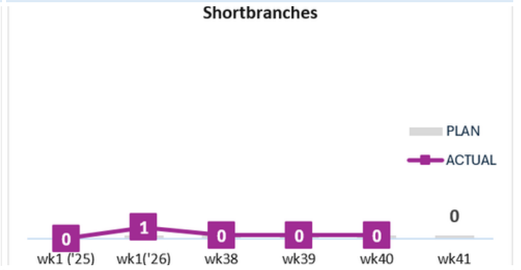
Furnace : 0 Jumpers

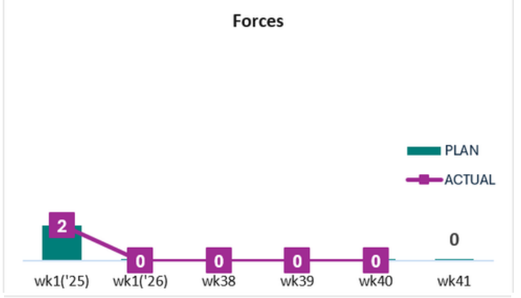


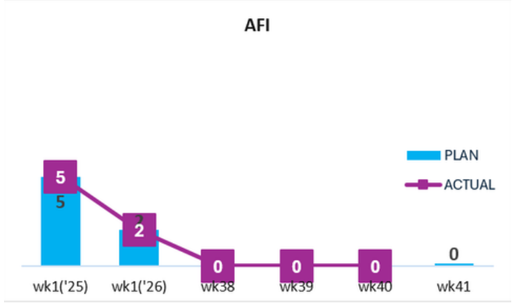




Converter : 0 Jumper







List of Remaining Jumper PP Area
Identify since 2026
without MOC = 3 Jumpers

AFI

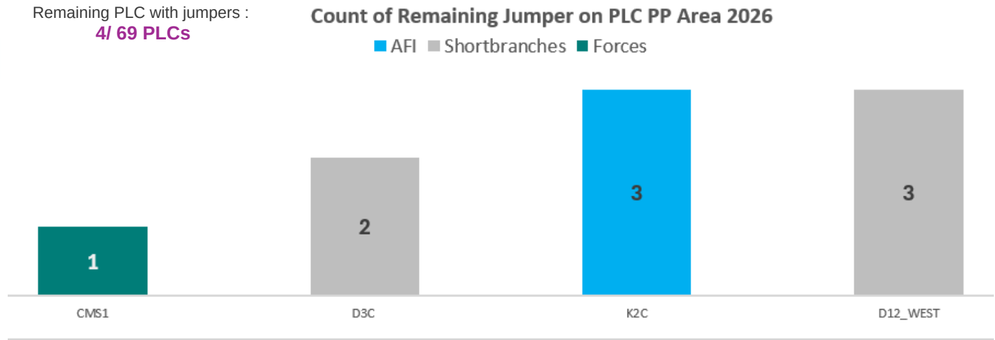
1. Conveying Blower opr ready E4AM01 on PLC K2C
2. Conveying Blower opr ready H21M03 on PLC K2C
3. Conveying Blower opr ready H21M04 on PLC K2C

TOTAL REMOVED JUMPER WK40 :
0 JUMPER

Total Jumper with Authorized/MoC 2026 : 3 Jumpers

PLC	Update Jumper	Installed	Auth. jumper	Due date
D3C	**1 Shortbranches** installed on PLC D3C, FSL (Flow Switch Low) Feed conv. Dryer 3	Wk10 2026	Exist	**Wk51 2026**
	1 Shortbranches installed on PLC D3C MainFlame On Interlock to TR ESP (need MOC to remove interlock)	Identify wk16	Exist	Need to confirm the due date.
CMS1	**1 Forces** installed on PLC CMS1 Level HiHi Switch Surge Bin - Digital_Alarms	Re-installed wk27	Exist	Wk27 2026
D12_ WEST	**1 Shortbranches** installed on PLC D12_WEST, KC221F4_8 Bag filter 1,3,4,5 is cleaning **1 Shortbranches** installed on PLC D12_WEST, G2M25 Running Forward **1 Shortbranches** installed on PLC D12_WEST, G2M24 Running Forward	Wk40 2026	Exist	Wk40 2026

Remaining PLC with jumpers :
4/ 69 PLCs



Reduction Kiln 4 Block Valve Sulphur Project
Process Plant

TYPE OF ACTIVITY		TECHNICAL SUPPORT	X	PROJECT		R&D

DESCRIPTION

Background

This project aims to integrate a PLC to enable control of the Sulphur BV and as trial for Optimize Sulphur Consumption initiative by utilizing the Sulphur Block Valve. The valve will be programmed to open and close automatically based on process interlocks, such as the operational status (open/close) of the Surge Bin Gate Valve. With the PLC implementation, the Block Valve will support both remote manual operation and automatic control via the HMI interface.

Facts and Findings

1. Sulphur Block Valve is currently operated via main drive hardwired control system and is not accessible through the HMI.
2. This project does not disrupt existing hardwire interlocks.

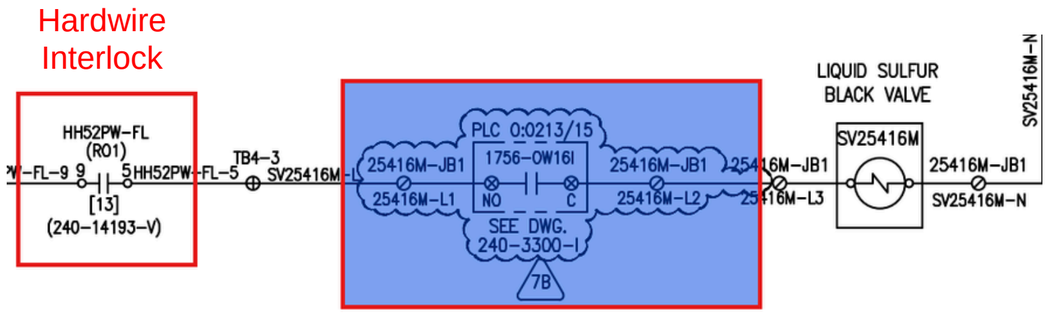
Action Plan

	Action	Who	When	Status
Short-Term	EWP Development	PPA	W27	DONE
	Construction (Cable Installation, Logic Development, HMI Development)	PPA & PPM	W35	DONE
	Commissioning	PPA & PPM	W40	DONE
Medium - Term	Monitor and Analyze Performance	PPA	W52	IP

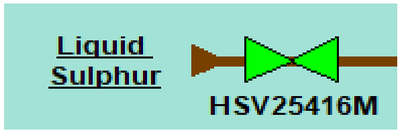
INFORMATION

Reported Date : September 30th, 2026
Area Location : Reduction Kiln 4

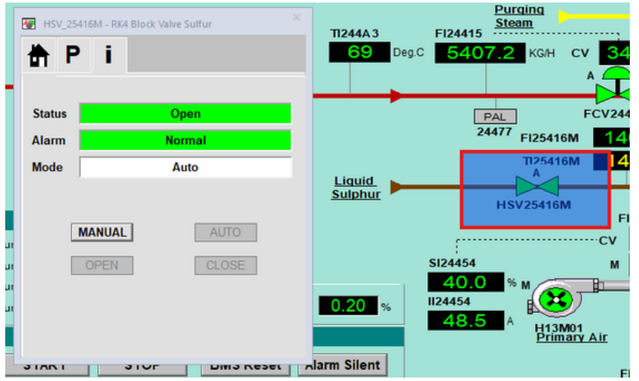
DOCUMENTATIC



BEFOR
E



AFTER



EF12 Trip & RK123 Standby 29-09-2026

TYPE OF ACTIVITY		TECHNICAL SUPPORT	X	PROJECT		R&D	

DESCRIPTION

Problem Statement:
At approximately 10:31-10:33 WITA on September 29th, 2026, EF1 & EF2 tripped simultaneously and RK1, RK2, and RK3 were secured via Emergency Stop following disturbances affecting the automation and control system, including partial HMI loss, PLC abnormalities, and network communication interruptions.

Findings :

1. At 10:31 WITA, the Process Plant Data Center HyperFlex Server power outage, resulting in a partial HMI blank condition on Process Plant Client B.
2. RK1, RK2, and RK3 were manually stopped via Emergency Stop by the operators to secure plant operation, although HMI Client A remained operational.
3. The following PLC power supply modules were found offline after the incident: PLC K1C IC K1-1PLC K1C IC K1-12 Chassis 04
4. The following PLC communication modules were found abnormal: EN2TR Module (PLC K2BMS) & CNBR Module (PLC K3 0 Chassis 02 & Chassis 08)
5. At 10:33 WITA, the F1-F2 communication switch experienced a momentary power interruption causing loss of network communication to PLC systems F1-F2 and K1-K2-K3.
6. EF1 and EF2 tripped simultaneously with multiple concurrent trip indications. Preliminary assessment suggests several signals may have been false trips generated following the power disturbance.

Preliminary Analysis :
Initial inspection indicates a momentary voltage dip on the UPS supply network. At the time of the incident, UPS #1 was under scheduled maintenance (fan replacement) and UPS #2 was supplying all critical loads. The disturbance is suspected to have affected sensitive devices, including PLC modules, network switches, HMI, and server infrastructure. Further investigation is ongoing through UPS event log review and PQM monitoring to determine the exact root cause.

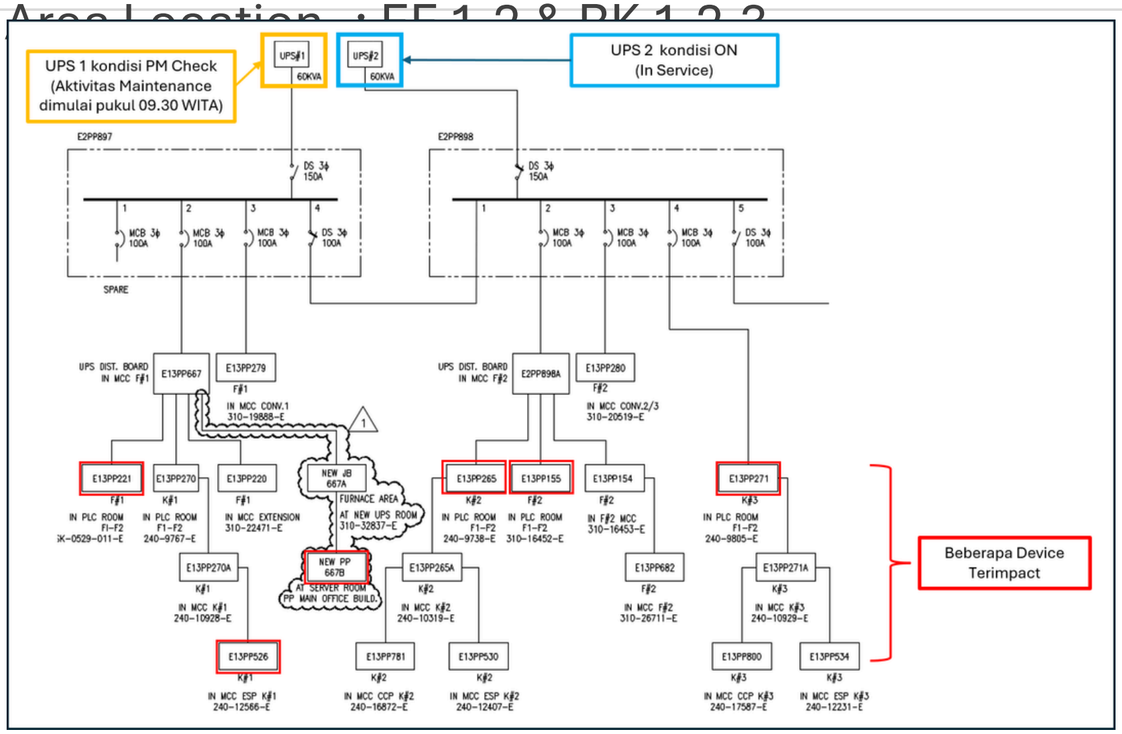
Current Condition:

1. UPS #1 and UPS #2 are operating normally and remain in service.
2. All critical control and automation loads are energized and operating normally.

INFORMATION

Reported Date : September 29th,2026

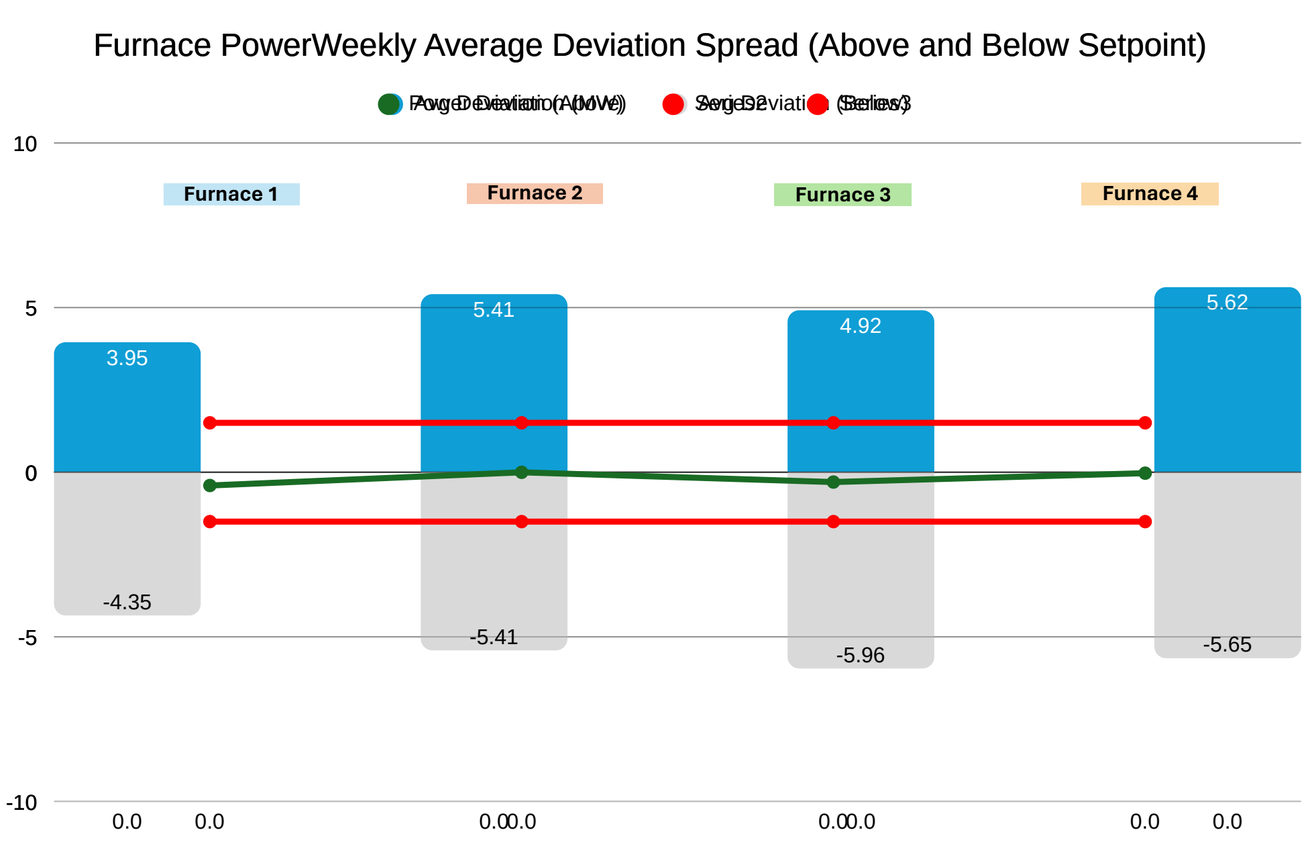
Area Location : EF 1,2 & RK 1,2,3



Next Step

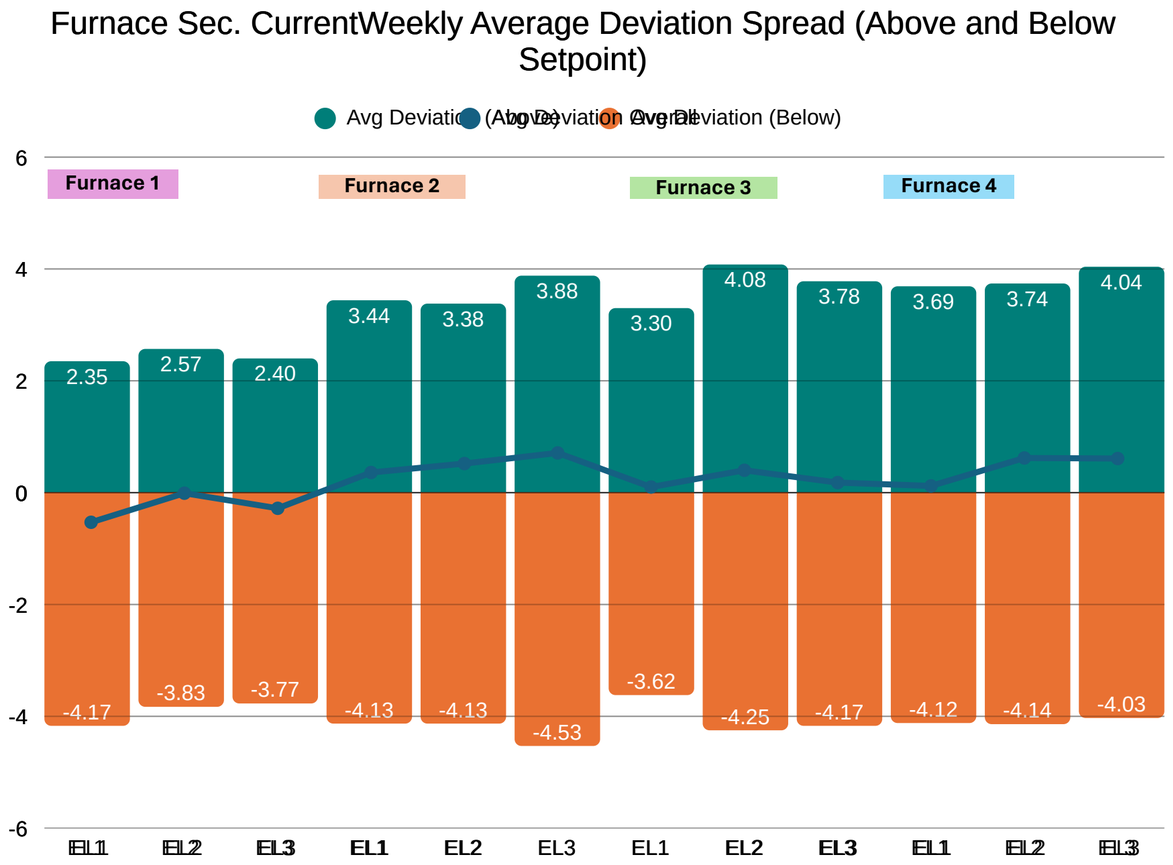
ACTION PLAN	DUE DATE	PROGRESS	STATUS
Collect Data UPS #1 & UPS #2 alarm/event records	WK.40	100%	Done
Discussion with Furnace Maintenance Team regarding UPS #1 PM activity	WK.40	100%	Done
Detailed UPS alarm history verification	WK.40	10%	I/P
Install PQM on UPS #1 & UPS #2 Incoming / Outgoing feeders	WK.40	0%	Open
Submit Final Automation Investigation Report	WK.40	0%	Open

PP Automation Update – Furnace Power



Furnace	Initial Interpretation	Recommendation
Furnace 1	Overall stable	-
Furnace 2	Overall stable	-
Furnace 3	Stable power after tuning process, but increased hydraulic oil temperature	• Tune Reduce electrode movement • perform speed test • Check hydraulic system performance
Furnace 4	Overall stable	-

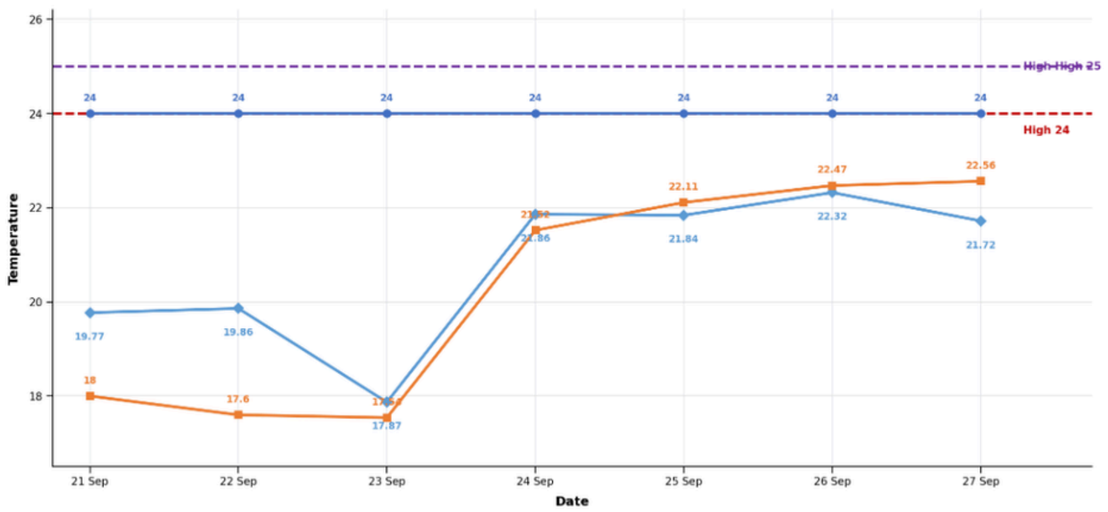
PP Automation Update – Sec. Current

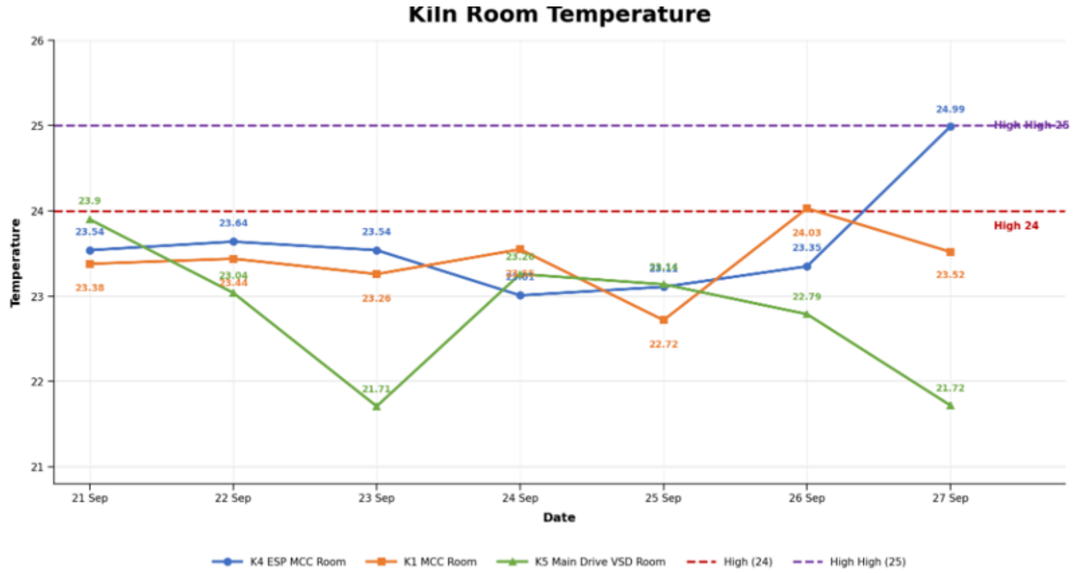


Furnace	Initial Interpretation	Recommendation
Furnace 1	Overall Stable	-
Furnace 2	Overall Stable	-
Furnace 3	Overall Stable	-
Furnace 4	Overall Stable	-

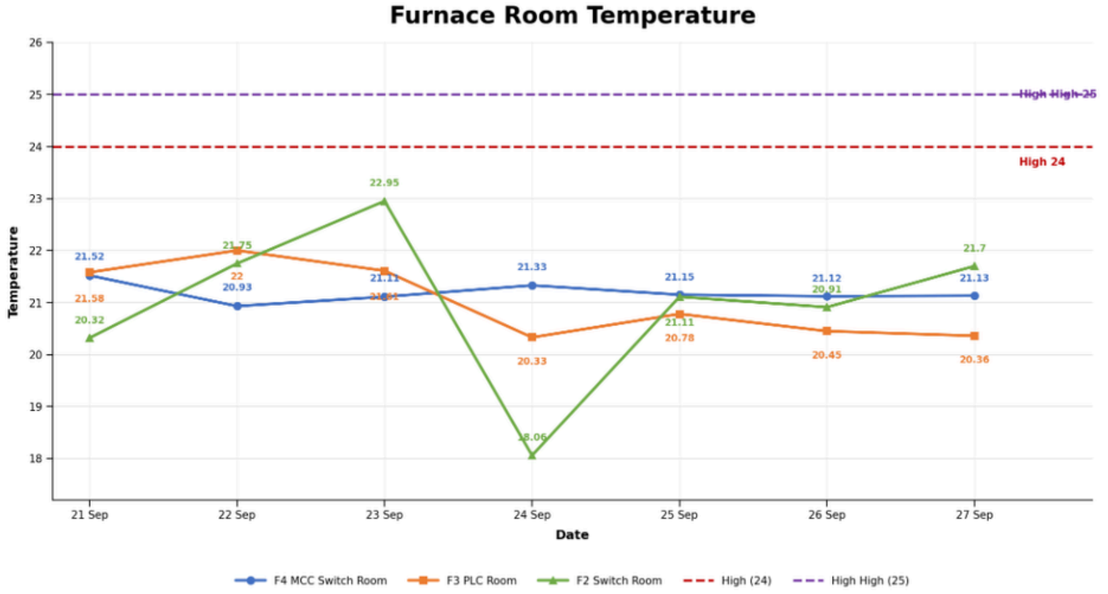
PLC ROOM TEMPERATURE MONITORING
Process Plant

DRYER ROOM NORMAL CONDITION

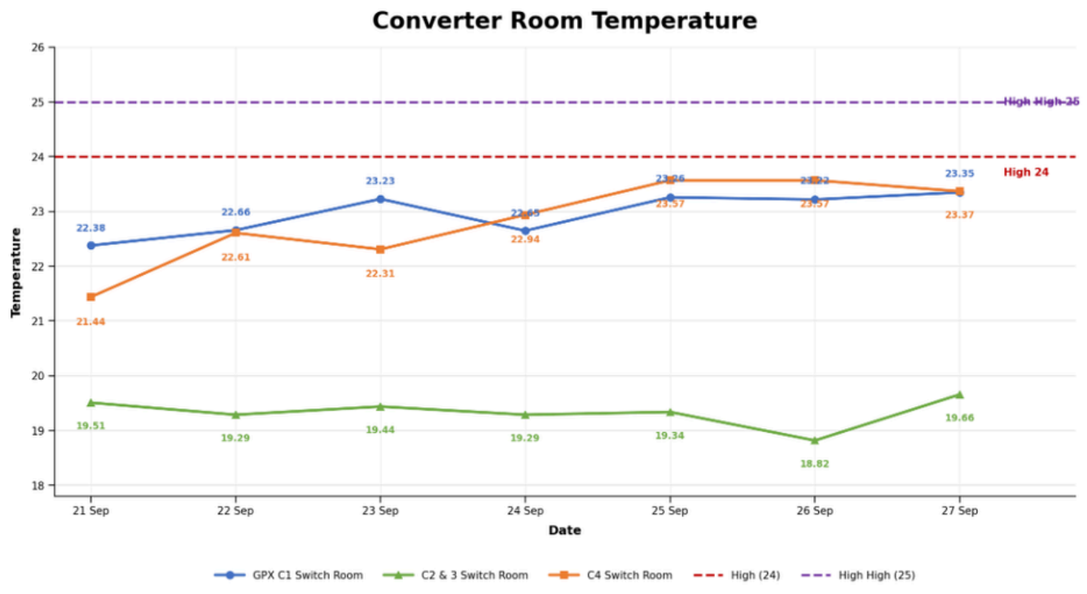




FURNACE ROOM NORMAL CONDITION



GPX C1 CONVERTER NORMAL CONDITION




MIND ID


VALE